Unit II: Electrochemistry and Concepts of Acid & Base:

Electrochemistry:

Definition, Electrochemical process, Faraday's laws of electrolysis; Oxidation, reduction, Redox reactions, electrode potential, electro chemical cell & electrolytic cell, Electro chemical series, Arrhenius theory of electrolytic dissociation, Electrolytic conductance-specific, equivalent and Molar, Ostwald's dilution law.

Concepts of Acid and base:

Bronsted, Arrhenius & Lewis theory, Concept of pH and pOH numerical, Buffer solutions, Indicators, Solubility product, Common ion effect with their applications.

Electrochemistry

Electrochemistry is the branch of chemistry that deals with the study of the relationship between electrical energy and chemical changes. Chemical reactions that involve the input or generation of electric currents are called electrochemical reactions. Such reactions are broadly classified into two categories that involve the transfer of electrons, typically between a solid electrode and an electrolyte, or between two different electrolytes. It includes processes like oxidation-reduction (redox) reactions, electrolysis etc.

"Electrochemistry deals with the chemical reactions produced by passing electric current through an electrolyte or the production of electric current through chemical reactions"

In simpler terms, electrochemistry focuses on how chemical reactions can produce electrical energy (like in batteries) or how electrical energy can drive chemical reactions (like in electroplating or the decomposition of water into hydrogen and oxygen).

Electrochemical process

An electrochemical process is a reaction that involves both electricity and chemical changes. In simple terms, it happens when a chemical reaction produces or uses electricity. There are two main types of electrochemical processes:

1. When electricity is used to drive a chemical reaction. This process is called electrolysis. For example, in electrolysis of water, electricity is used to split water (H_2O) into hydrogen gas (H_2) and oxygen gas (O_2).
2. When chemical reactions generates electric energy. A common example is a battery, where chemical reactions inside the battery produce electricity that powers devices like your phone or flashlight.

In both cases, there are two important things that happen:

- **Electron flow**: Electrons move from one part of the system to another, creating an electric current.
- **Chemical change**: Substances at the electrodes (the parts where electricity enters or leaves the system) undergo chemical reactions.

To summarize, electrochemical processes is conversion between chemical energy and electrical energy.

Faraday's Law of Electrolysis:

Faraday's laws of electrolysis are fundamental principles that govern the process of electrolysis, where electrical energy is used to drive a chemical reaction. These laws are named after Michael Faraday, who formulated them in the 19th century. There are two laws of electrolysis:

1. Faraday's First Law of Electrolysis:

This law states that the amount of substance deposited or liberated at an electrode during electrolysis is directly proportional to the amount of electric charge passed through the electrolyte.
In mathematical terms:

$$m \alpha Q$$
$$m = Z.Q$$
$$m = Z.i.t \qquad (Q = i.t)$$

Where:

- m = mass of a substance (in grams, g) deposited or liberated at an electrode
- Q = amount of charge (in coulombs, C)
- i = electricity passed
- t = time taken
- Z is the proportionality constant. Its unit is grams per coulomb (g/C). It is also called the electrochemical equivalent. Z is the mass of a substance deposited at electrodes during electrolysis by passing 1 coulomb of charge.

In simple terms, if you pass a greater amount of electric charge through the electrolyte, more of the substance will be deposited at the electrodes.

2. Faraday's Second Law of Electrolysis:

This law states that the amounts of different substances deposited or liberated at the electrodes by the same quantity of electricity are proportional to their equivalent masses (which depend on the molar mass and valency of the substances).
In simpler terms:

$$\frac{m1}{m2} = \frac{E1}{E2}$$

Where:

- m_1 and m_2 are the masses of the substances deposited or liberated at the electrodes,

- E_1 and E_2 are the equivalent masses

The equivalent weight or chemical equivalent of a substance can be defined as the ratio of its atomic weight and valency.

(Equivalent weight = Atomic weight/Valency)

This law implies that if the same amount of electric charge for two different substances, the substance with the lower equivalent mass will be deposited in a lesser quantity.

On combining first and second law:

$$\frac{Z1Q}{Z2Q} = \frac{E1}{E2}$$

$$\frac{Z1}{Z2} = \frac{E1}{E2}$$

$$\frac{E1}{Z1} = \frac{E2}{Z2}$$

$$\frac{E}{Z} = F$$

Where F, is faraday constant. Value of F is 96500 coulomb.

Then

$$Z = E / 96500$$

If we pass 1 Faraday (96500) of charge in an electrolytic cell, then 1gm of the equivalent weight of the substance will get deposited.

Summary:

- Faraday's First Law: The mass of a substance deposited or liberated is proportional to the electric charge.
- Faraday's Second Law: The amounts of different substances deposited by the same charge are proportional to their equivalent masses.

These laws help to calculate the outcomes of electrolysis processes in terms of the amount of material involved based on the amount of electric current and time.

Redox reactions

Redox reactions, short for **oxidation-reduction reactions**, are a type of chemical reaction that involves the transfer of electrons between two substances. In these reactions, one substance loses electrons (oxidation), while another gains electrons (reduction).

1. Oxidation:

Oxidation is one half of a redox reaction (the other being reduction). It refers to the process in which an atom, ion, or molecule loses electrons. When something is oxidized, its oxidation state increases (it becomes more positive).

Key Points about Oxidation:

- **Electron loss**: Oxidation **i**s the process in which an atom, ion, or molecule loses electrons**.**
- In organic chemistry, Oxidation often involves adding oxygen or removing hydrogen.
- **Oxidation state increases**: As electrons are lost, the oxidation state of the atom or ion becomes more positive.
- The substance that loses electrons is said to be oxidized.
- **Oxidizing agent (Oxidant)**: The substance that accepts the electrons and causes the oxidation of another substance is called the oxidizing agent**.**

Example:

In the reaction between zinc and copper sulphate ($CuSO_4$):

$$Zn\ (s) + CuSO_4 \rightarrow Cu\ (s) + ZnSO_4$$

Oxidation Half*:*

$Zn(s) \rightarrow Zn^{2+}\ (aq) + 2e^-$

Here, zinc (Zn) loses two electrons, so it's oxidized, and its oxidation state goes from 0 to +2.

2. Reduction:

Reduction is the opposite of oxidation and refers to the process where an atom, ion, or molecule gains electrons**.** When a substance is reduced, its oxidation state decreases (i.e., it becomes more negative). In simpler terms, reduction is the process that removes oxygen or adds hydrogen to a compound.

- Reduction is the process in which an atom, ion, or molecule gains electrons**.**
- The substance that gains electrons is said to be reduced**.**
- During reduction, the oxidation state of the element decreases (i.e., it becomes more negative).
- **Reducing agent (Reductant)**: The substance that loses electrons and causes the reduction of another substance is called the reducing agent**.**

Example:

In the same reaction ($Zn\ (s) + CuSO_4 \rightarrow Cu\ (s) + ZnSO_4$):

Reduction Half:

$Cu^{2+}\ (aq) + 2e^- \rightarrow Cu(s)$

Copper ions (Cu^{2+}) gain two electrons and get reduced to form solid copper (Cu).

3. Oxidizing Agent and Reducing Agent:

- **Oxidizing agent**: The substance that gains electrons and is reduced itself. It causes oxidation to happen. In the zinc and copper sulphate reaction, Cu^{2+} is the oxidizing agent.
- **Reducing agent**: The substance that loses electrons and is oxidized itself. It causes reduction to happen. In the same reaction, zinc (Zn) is the reducing agent.

4. Electron Transfer:

In a redox reaction, the total number of electrons lost during oxidation is always equal to the total number of electrons gained during reduction. This ensures that the overall charge is balanced.

Electrochemical Cells

A spontaneous chemical reaction is one that can occur on its own, and in such a reaction, the system's Gibbs energy falls. This energy is then transformed into electrical energy. It is also feasible to force non-spontaneous processes to occur by providing external energy in the form of electrical energy. Electrochemical Cells are used to carry out these interconversions.

Types

Two types of electrochemical cells are present: **Galvanic cells**, which converts chemical energy into electrical energy and **electrolytic cells** which converts electrical energy into chemical energy.

1. **Galvanic Cells**

A spontaneous chemical process or reaction is used to extract cell energy, which is then transformed to electric current.

For example, a Daniell Cell is a Galvanic Cell in which the redox reaction is carried out using Zinc, Copper electrode and solution of their salt.

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

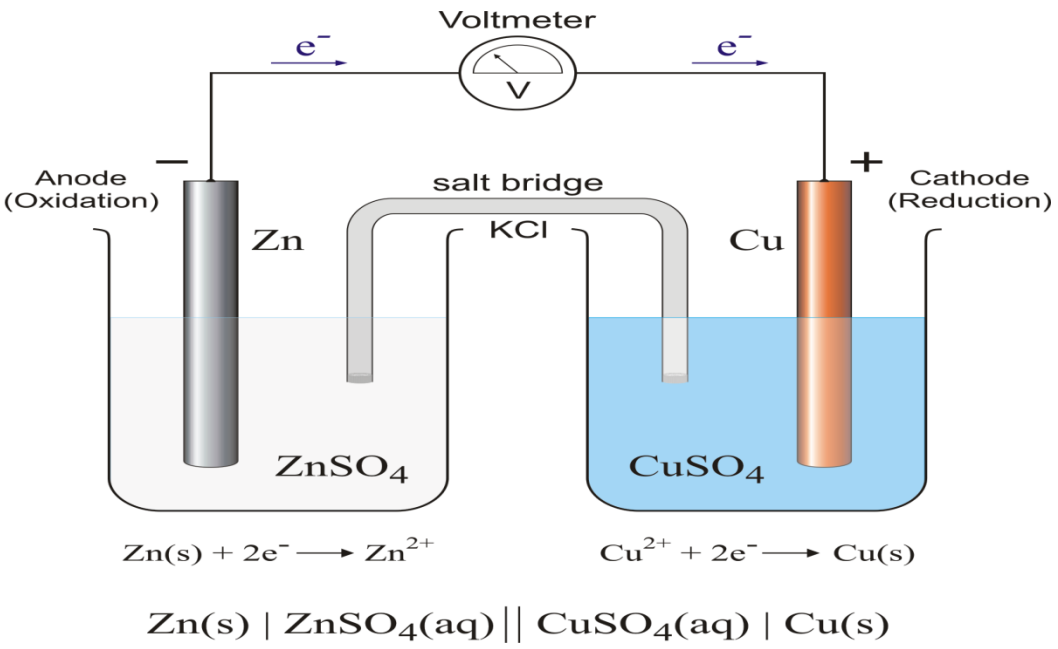
Oxidation Half*: (at the anode)*

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

Reduction Half: *(at the cathode)*

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

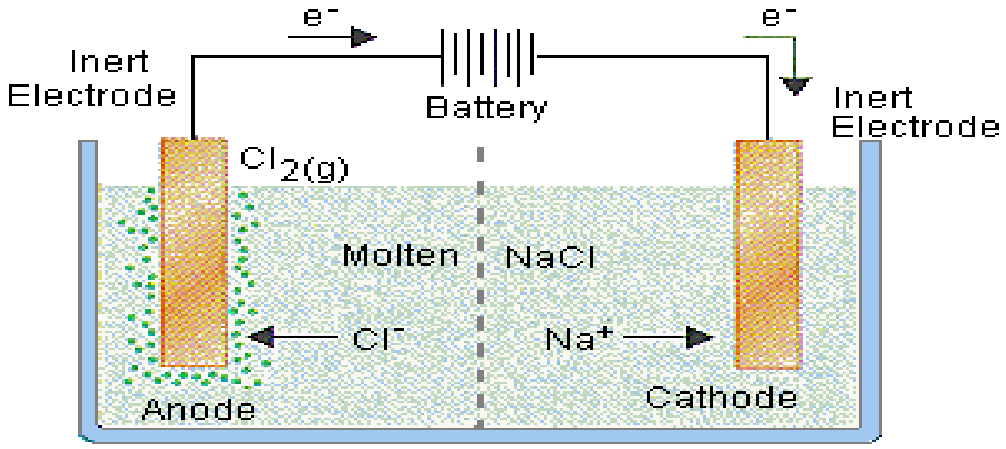
The reducing agent is Zn, and the oxidising agent is Cu^{2+}. Electrodes are another name for half cells. The anode is the oxidation half, and Cathode is the reduction half. Cathode is a term used to describe a type of electrode. In the external circuit, electrons pass from anode to cathode. Negative polarity is assigned to the anode. Positive polarity is assigned to the cathode. Daniell Cell is a fictional character created by Daniell Cell. The anode is Zn, while the cathode is Cu.



Galvanic Cell

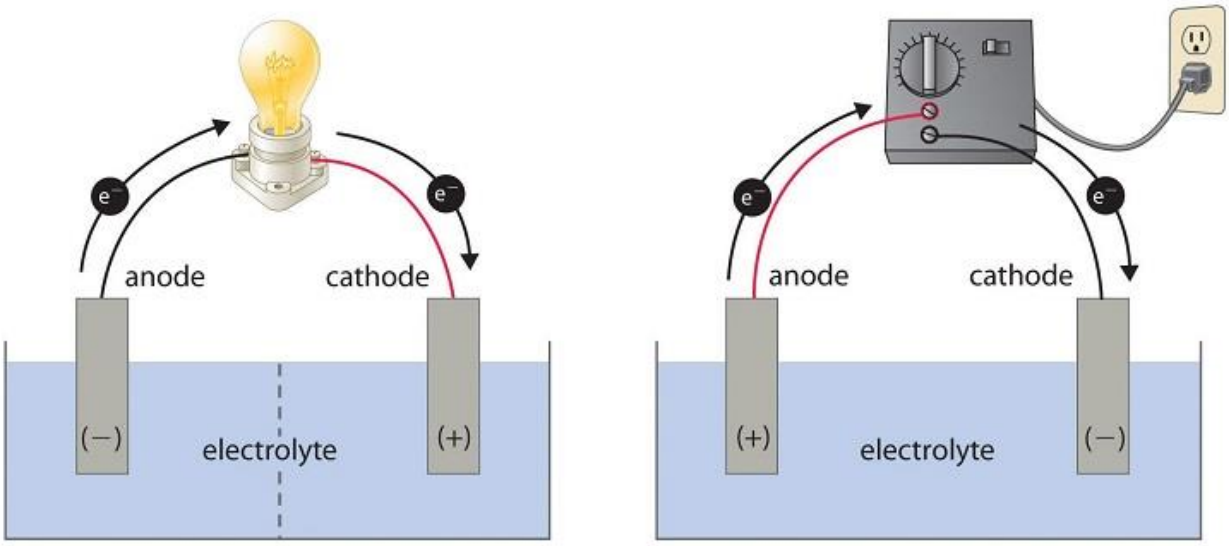
2. Electrolytic Cell

These electrodes are submerged in an electrolytic solution that contains both cations and anions. When current is supplied, the ions migrate towards electrodes of opposite polarity, where they undergo simultaneous reduction and oxidation.



Electrolytic Cell

Comparison between Galvanic Cell & Electrolytic Cell



GALVANIC CELL	ELECTROLYTIC CELL
Energy released by spontaneous redox reaction is converted to electrical energy.	Electrical energy is used to drive nonspontaneous redox reaction.
Oxidation half-reaction: $Y \rightarrow Y^+ + e^-$	Oxidation half-reaction: $Z^- \rightarrow Z + e^-$
Reduction half-reaction: $Z^+ + e^- \rightarrow Z$	Reduction half-reaction: $Y^+ + e^- \rightarrow Y$
Overall cell reaction: $Y + Z \rightarrow Y^+ + Z^-$ $(G < 0)$	Overall cell reaction: $Y^+ + Z^- \rightarrow Y + Z$ $(G > 0)$

Preferential Discharge of Ions

When more than one cation or anion is present, the discharge process becomes competitive. If more than one type of ion is attracted towards a particular electrode, then the one discharged is the ion that requires the least energy or ions that have the lower discharge potential.

Electrode Potential

It can be defined as an element's tendency to lose or gain electrons when in contact with its own ions, causing it to become positively or negatively charged. Depending on whether oxidation or reduction has occurred, the electrode potential will be referred to as oxidation or reduction potential. It is denoted by the sign E. It is also called single electrode potential. It is not possible to measure accurately the absolute value of single electrode potential directly. Only the difference in potential between two electrodes can be measured experimentally. So, in an

experiment to measure electrode potential, in a cell one electrode is used as a reference electrode whose potential is already known and another electrode is used of unknown potential.

$$M(s) \xrightarrow{\text{Oxidation}} M^{n+}(aq) + ne^-$$

$$M^{n+}(aq) + ne^- \xrightarrow{\text{Reduction}} M(s)$$

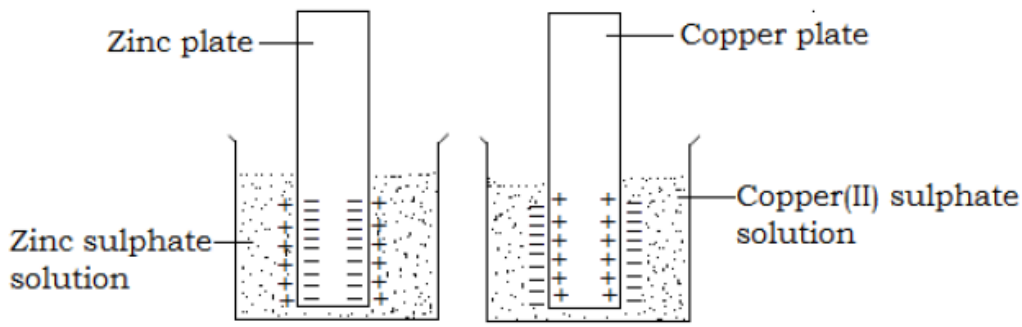
Characteristics

(a) The magnitude of the oxidation and reduction potentials is equal.

(b) E is not a thermodynamic property, its values do not add up.

Factors affecting electrode potential

Electrode potential depends on the nature of the metal, temperature and the concentration of metal ions in solution.



In such systems, there are to tendencies:

i) The metal (zinc) tends to dissolve to form ions. The electrons remain behind on the metal surface (zinc surface) and so it becomes negatively charged because of the excess electrons.

Equation

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

So, there exists a potential difference between the metal and its solution and this is called the **electrode potential.** The electrode potential measures the reactivity of the metal.

ii) The metal ions (copper ions) gain electrons and get deposited on the metal surface.

Equation

$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

Standard Electrode Potential (E^o)

It can be described as an electrode's electrode potential measured in comparison to a standard hydrogen electrode under standard conditions. Standard electrode potential is denoted by E^0. The following are the standard conditions:

(i) A 1M concentration of each ion in the solution.

(ii) A 298 K temperature.

(iii) Each gas has a pressure of one bar.

Standard hydrogen electrode is a gas – ion electrode. It is used as a reference electrode for determination of standard electrode potential of elements and other half cells. It can act as anode half - cell as well as cathode half-cell. Value of its standard reduction potential and standard oxidation potential is always zero at 25℃ or 298K.

Electrochemical Series

Electrochemical series, also referred to as activity series, is a list that describes the arrangement of elements in the order of their increasing standard electrode potential (Reduction Potential) values or decreasing reactivity.

Each half-reaction is written as a reduction reaction, and the standard electrode potential is measured under standard conditions (298 K, 1 M concentration for solutions, and 1 atm pressure for gases).

The standard electrode potential is measured in volts (V) relative to the standard hydrogen electrode (SHE), which is defined as having a potential of 0 V at all temperatures.

Key Concepts:

- **Reduction**: The gain of electrons (a substance with a higher (more positive) electrode potential tends to be reduced more easily).
- **Oxidation**: The loss of electrons (a substance with a lower (more negative) electrode potential tends to undergo oxidation more easily).
- **Standard electrode potential (E°)**: The voltage associated with a half-reaction at standard conditions.

Applications of the Electrochemical Series

1. **Prediction of Reaction Direction**:
    - The electrochemical series helps predict whether a specific redox reaction will occur. A reaction is spontaneous if the reduction half-reaction (at the cathode) has a more positive electrode potential than the oxidation half-reaction (at the anode).

- For example, if zinc (E° = -0.76 V) is paired with copper (E° = +0.34 V) in a galvanic cell, zinc will undergo oxidation (since its electrode potential is lower), and copper will undergo reduction (since its electrode potential is higher).

2. **Design of Galvanic Cells (Batteries)**:
   - The electrochemical series is crucial in designing electrochemical cells like galvanic cells and batteries. By selecting materials with appropriate electrode potentials, engineers can maximize the voltage and efficiency of the cells. For instance, in the Daniell cell, zinc is oxidized, and copper is reduced, providing a voltage of 1.10 V.
3. **Corrosion Protection**:
   - The electrochemical series is used to understand and prevent corrosion. The metal with a lower electrode potential will tend to corrode (oxidize), while a metal with a higher electrode potential will resist corrosion.
   - **Galvanization** (coating iron with zinc) works because zinc, with a lower potential, sacrifices itself to protect iron.
4. **Electroplating and Electrorefining**:
   - Electroplating involves depositing a metal onto a surface by using a reduction reaction. The electrochemical series helps determine which metal will deposit at the cathode when an electric current is passed through a solution of metal ions.
   - In **electrorefining**, metals like copper are purified by using the electrochemical series to selectively reduce impurities and leave the pure metal behind.
5. **Determining Cell Potential**:
   - The overall cell potential of a galvanic cell can be calculated by subtracting the standard reduction potentials of the two half-reactions. A positive indicates a spontaneous reaction.
6. **Electrolysis of Water**: The electrochemical series helps in understanding the electrolysis process. The standard electrode potential for water splitting is relatively negative, meaning it requires a certain voltage to break water into hydrogen and oxygen.
7. **Redox Titrations**:
   - The electrochemical series aids in performing redox titrations, where the reduction potentials of the involved species help determine the endpoint of the titration.

Cell Potential or EMF of a Cell

Cell potential is the difference between the electrode potentials of two half cells. If no current is pulled from the cell, it is known as electromotive force (EMF).

$$E_{cell} = E_{cathode} + E_{anode}$$

For this equation we take oxidation potential of anode and reduction potential ofcathode.

Since anode is put on left and cathode on right, it follows therefore:

$$= E_R + E_L$$

For a Daniel Cell, therefore:

$$E^o_{cell} = E^o_{Cu^{2+}/Cu} + E^o_{Zn/Zn^{2+}} = 0.34 + 0.76 = 1.10V$$

Cell Diagram or Representation of a Cell

In accordance with IUPAC recommendations, the following conventions or notations are used to write the cell diagram. The Daniel cell has the following representation:

$$Zn(s) \mid Zn^{2+}(C) \mid\mid Cu^{2+}(C) \mid Cu(s)$$

(a) The anode half cell is written on the left, while the cathode half cell iswritten on the right.

(b) The metal is separated from an aqueous solution of its own ions by a single vertical line

Anodic Chamber: $Zn(s) \mid Zn^{2+}(aq)$

Cathodic Chamber: $Cu^{2+}(aq) \mid Cu(s)$

(a) A salt bridge is represented by a double vertical line.

(b) After the formula of the corresponding ion, the molar concentration (C) is placed in brackets.

The cell's e.m.f. value is written on the cell's extreme right side. As anexample:

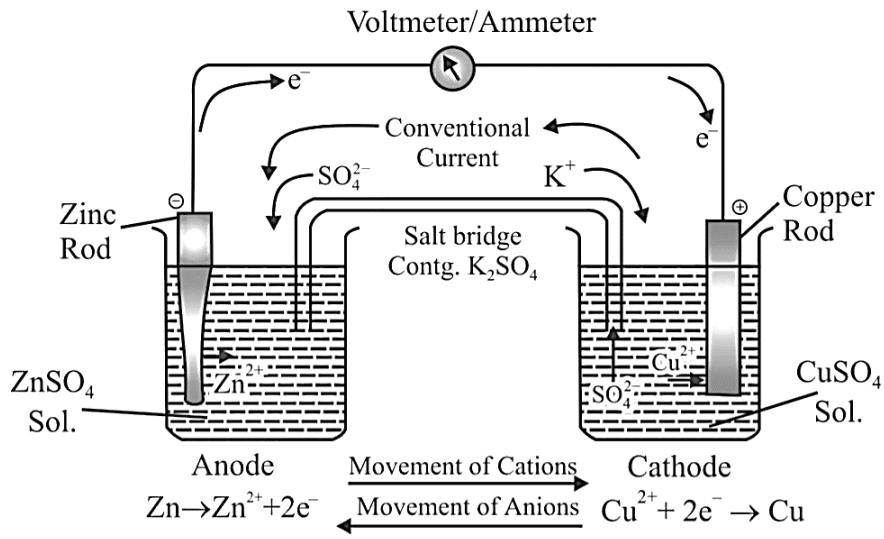
$$Zn(s) \mid Zn^{2+}(1M) \mid\mid Cu^{2+}(1M) \mid Cu \text{ , EMF = +1.1 V}$$

If an inert electrode, such as platinum, is used in the cell's construction, it may be written in brackets alongside the working electrode, as when a zinc anode is coupled to a hydrogen electrode.

$$Zn(s) \mid Zn^{2+}(C_1) \mid\mid H^+(C_2) \mid H_2(Pt)(s)$$

Salt Bridge

The salt bridge maintains charge balance and completes the circuit by allowing ions to flow freely through it. It contains a gel containing an inert electrolyte such as Na_2SO_4 or KNO_3. Through the salt bridge, negative ions travel to the anode and positive ions flow to the cathode, maintaining charge balance and allowing the cell to function.



Electrochemical cell based on the redox reaction,
$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

Spontaneity of a Reaction

$\Delta G = -nFE_{cell}$

ΔG should be negative and cell potential should be positive for a spontaneous cell reaction.

In the following equation, if we take the standard value of cell potential, we will also get the standard value of ΔG .

The Arrhenius theory of electrolytic dissociation

The Arrhenius theory of electrolytic dissociation, proposed by Swedish chemist Svante Arrhenius in 1887, explains how electrolytes dissociate into ions when dissolved in water, allowing the solution to conduct electricity.

According to this theory, when an electrolyte (a substance that can conduct electricity) dissolves in water, it splits into positively charged ions (cations) and negatively charged ions (anions). These ions are free to move in the solution, and it is this movement of charged particles that enables the flow of electric current.

1. **Electrolytes**: These are substances that dissociate into ions when dissolved in water, making the solution conductive. An electrolyte can be an acid, base or salt.

    - **Strong electrolyte:** This electrolyte dissociates completely into ions though it is

in a concentrated solution. For example, NaOH, KOH, HCl, NaCl, etc. Most of the salts are always strong electrolytes.

$$NaCl \rightarrow Na^+ + Cl^-$$

Sodium chloride

- **Weak electrolyte:** This electrolyte dissociates only to a small extent in a solution. For example, CH_3COOH, NH_4OH, etc.

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

Acetic acid

2. **Dissociation**: When an electrolyte dissolves, the solvent (water) interacts with the molecules of the electrolyte, pulling them apart into individual ions. For example, when NaCl (sodium chloride) dissolves in water, it dissociates into Na^+ (sodium ions) and Cl^- (chloride ions).

Degree of Ionisation or Dissociation

The degree of dissociation is calculated by the ratio of the total number of molecules which is ionised at the equilibrium state.

$$\alpha = \frac{Number\ of\ moles\ ionised}{Total\ number\ of\ moles}$$

3. **Conductivity**: The dissociated ions are responsible for carrying the electric current through the solution. The more ions present, the better the conductivity.

Arrhenius's theory was fundamental in understanding the behavior of electrolytes and paved the way for later developments in physical chemistry and the study of ionization.

Electrolytic conductance

Electrolytic conductance (or simply conductance) refers to the ability of an electrolyte solution to conduct electricity, and it’s a measure of how easily ions move through the solution. The concept is linked to the ions produced by electrolytic dissociation, which carry electrical charge through the solution.

Specific conductance

Specific Conductance is the ability of a substance to conduct electricity. It is the reciprocal of specific resistance (ρ). It is also known as Conductivity. Conductance means the flow of current through the liquid conductor. Specific conductance is defined as the conducting capacity of a solution of the dissolved electrolyte and the whole solution is being placed between two electrodes are 1 cm^2 and length 1 cm.

1. **Definition**: Specific conductance (denoted by κ or σ) is the conductance of a unit volume of the solution. The higher the concentration of ions and their mobility, the higher the specific conductance.
2. **Relation to Ion Concentration**: Specific conductance increases as the concentration of ions in solution increases, because more ions are available to carry the electric current. However, the relationship is not always linear for weak electrolytes because they may not dissociate completely.
3. **Units of Specific Conductance**: Unit is Ω^{-1} cm^{-1} or mho cm^{-1}. In SI units, it is measured in siemens per meter (S/m) .

Effect of dilution and temperature on specific conductance

The value of specific conductance depends on the number of ions per unit volume of solution. On dilution specific conductance of a solution decreases because the number of ions per unit volume decreases. The lower the number of ions per unit volume, the lower the value of conductivity. Therefore, the specific conductance value increases with an increase in concentration.

In electrolytic conductors, the ions are charge carriers and with increase in temperature, ionization increases and hence, conductivity increases.

Equivalent Conductance: Equivalent Conductance is defined as the conductance of all the ions produced by dissolving one gram equivalent of an electrolyte in solution.It is represented by λ_{eq}. Its units are $\Omega^{-1}cm^2equiv^{-1}$ or mho cm^2equiv^{-1} or S m^2equiv^{-1}.

Molar Conductance: It refers to the conductance of a solution containing one mole of electrolyte in a given volume. It is represented by Λ_m. Its units are $\Omega^{-1}cm^2mol^{-1}$ or mho cm^2mol^{-1} or S m^2 mol^{-1}. For strong electrolytes, molar conductance increases with dilution

(as ion-ion interactions decrease), while for weak electrolytes, the molar conductance increases with dilution due to greater dissociation.

Effect of dilution and temperature on Equivalent and Molar conductance

On dilution, the equivalent conductance and the molar conductance increase. This is because the degree of dissociation and the total number of ions increases. On increasing the temperature of an electrolyte solution ionization increased, which results in a higher concentration of ions and consequently an increase in equivalent and molar conductance.

Formula:

- **Specific Conductance (κ)**:

$$\kappa = \frac{1}{\rho}$$

$$\kappa = \frac{l}{R.A} \qquad \left(R = \rho.\frac{l}{A} \quad \text{or} \quad \rho = \frac{R.A}{l} \right)$$

κ = conductance x cell constant (conductance = 1/R, cell constant = l/A)

Where:

- *R* is the resistance (in ohms),
- ρ is specific resistance
- l *is* the distance between the two electrodes placed in electrolyte
- *A* is the area of the cross-section

- **Equivalent Conductance (λ_{eq})**:

$$\lambda_{eq} = \kappa \text{ x } V$$

V is volume (mL) containing 1gm equivalent electrolyte

Also

$$\lambda_{eq} = \frac{\kappa \, x \, 1000}{N}$$

N is normality of solution.

- **Molar Conductance (Λ_m):**

$$\Lambda_m = \kappa \text{ x } V$$

V is volume (mL) containing 1mole of electrolyte.

Also

$$\Lambda_M = \frac{\kappa \; x \; 1000}{M}$$

M is molarity of solution.

Example:

When NaCl dissolves in water, it dissociates completely into Na^+ and Cl^- ions. If we have a higher concentration of NaCl, the number of ions in solution increases, and so does the specific conductance. However, for very dilute solutions, the ions are far apart, and their mobility increases, so molar conductance increases as well.

Ostwald's dilution law

Ostwald's dilution law relates the dissociation constant of the weak electrolyte with the degree of dissociation and the concentration of the weak electrolyte.

The law is expressed as:

$$\mathbf{K_d} = \frac{\boldsymbol{\alpha 2 \; x \; c}}{(\mathbf{1} - \boldsymbol{\alpha})}$$

If α is too small, (1- **α) =** 1 then

$$\mathbf{K_d = \alpha^2 C}$$

Where:

- **α** is the degree of dissociation (the fraction of the electrolyte that dissociates).
- **K_d** is the dissociation constant.
- **C** is the molar concentration of the electrolyte.

1. The law is applicable to weak electrolytes, which don't completely dissociate in solution (unlike strong electrolytes).

2. As the concentration (C) of the electrolyte decreases (i.e., when the solution becomes more dilute), the degree of dissociation (α) increases.
3. Ostwald's law helps to explain the behavior of weak acids, bases, or salts in dilute solutions.

In simpler terms, it tells us that for weak electrolytes, the more dilute the solution, the more the electrolyte will dissociate. However, the extent of dissociation is also governed by the intrinsic dissociation constant.

The Arrhenius concept of acids and bases:

The Arrhenius concept of acids and bases, introduced by Swedish chemist Svante Arrhenius in the late 19th century, defines acids and bases in terms of their behavior in water. This concept is one of the earliest and most basic theories for understanding acids and bases.

Arrhenius Acid:

An Arrhenius acid is a substance that increases the concentration of hydrogen ions (H^+) or protons in aqueous solutions. When dissolved in water, the acid dissociates to release H^+ ions. These acids have their characteristic properties (e.g., sour taste, ability to turn blue litmus paper red).

- **Example**:

  Hydrochloric acid (HCl) dissociates in water as follows:

  $HCl \rightarrow H^+ + Cl^-$

  Here, H^+ ions are released into the solution, making it acidic.

Arrhenius Base

An Arrhenius base is a substance that increases the concentration of hydroxide ions (OH^-) in aqueous solutions. When a base dissolves in water, it dissociates to release OH^- ions. These ions are responsible for the basic (alkaline) properties of the solution, such as a bitter taste and the ability to turn red litmus paper blue.

- **Example**:

  Sodium hydroxide (NaOH) dissociates in water as follows:

  $NaOH \rightarrow Na^+ + OH^-$

  Here, OH^- ions are released, making the solution basic.

Key Points of the Arrhenius Concept:

1. **Acids**: Increase the concentration of H^+ ions in water.
2. **Bases**: Increase the concentration of OH^- ions in water.
3. The concept only applies to aqueous solutions (solutions in water).

Limitations:

While the Arrhenius concept is helpful for many simple reactions, it has some limitations:

- It only applies to aqueous solutions and cannot account for acid-base reactions in non-aqueous solvents.
- It doesn't explain reactions where acids or bases do not directly release H^+ or OH^- ions, like ammonia (NH_3) in water, which is a base but does not directly release OH^-. Instead, it reacts with water to form OH^- ions.

Despite these limitations, the Arrhenius concept is a foundational principle in understanding acid-base chemistry.

The Brønsted-Lowry concept of acids and bases

The Brønsted-Lowry concept of acids and bases, introduced by Johannes Nicolaus Brønsted and Thomas Lowry in 1923, defines acids and bases based on their ability to donate or accept protons (hydrogen ions, H^+).

Here are the key points of the Brønsted-Lowry theory:

1. Brønsted-Lowry Acid:

An acid is a substance that **donates a proton (H^+)**. When an acid dissociates, it releases a proton, which can then interact with a base.

- Example: In the reaction of hydrochloric acid (HCl) with water:

  $HCl \rightarrow H^+ + Cl^-$

  Here, **HCl** donates a proton to the water molecule, making HCl a Brønsted-Lowry acid.

2. Brønsted-Lowry Base:

A base is a substance that **accepts a proton (H^+)**. Bases can accept a proton from an acid to form a conjugate acid.

- Example: In the reaction between ammonia (NH_3) and water:

  $NH_3 + H_2O \rightarrow NH_4^+ + OH^-$

  Here, **NH_3** accepts a proton from the water, making it a Brønsted-Lowry base.

3. Conjugate Acid-Base Pairs:

- When an acid donates a proton, the substance that remains after the proton is lost is called the **conjugate base**.
- When a base accepts a proton, the substance formed is called the **conjugate acid**.

For example:

- In the reaction of acetic acid (CH_3COOH) with water:

  $CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$

  Here, CH_3COOH (acetic acid) is the Brønsted-Lowry acid, and it donates a proton to water. The conjugate base is CH_3COO^-, and the conjugate acid is H_3O^+ (hydronium ion).

4. Amphiprotic Substances:

- Some substances can act as both acids and bases, depending on the situation. These substances are called **amphiprotic**. Water (H_2O) is a classic example. It can act as an acid by donating a proton, or as a base by accepting a proton.
  - As an acid: $H_2O \rightarrow H^+ + OH^-$
  - As a base: $H_2O^+ + H^+ \rightarrow H_3O^+$

This theory is more general than the Lewis concept, which involves electron pairs rather than protons, and is widely used to explain acid-base reactions in aqueous solutions.

The Lewis theory of acids and bases

The Lewis theory of acids and bases is an extension of the traditional Brønsted-Lowry and Arrhenius theories. Instead of focusing on protons (H^+) and hydroxide ions (OH^-), the Lewis theory defines acids and bases in terms of electron pairs**.**

Lewis Acid:

A Lewis acid is a substance that can accept a pair of electrons from another substance (it is an electron pair acceptor). This occurs because Lewis acids have an incomplete octet or a vacant orbital that allows them to accept electron pairs.

Examples:

- **Boron trifluoride (BF_3):** Boron in BF_3 has only six electrons in its valence shell, making it electron-deficient and capable of accepting an electron pair.
- **Aluminum chloride ($AlCl_3$):** Similar to BF_3, $AlCl_3$ has an incomplete octet and can accept electrons.

Lewis Base:

A Lewis base is a substance that can donate a pair of electrons to another substance (it is an electron pair donor). Lewis bases have lone pairs of electrons that they can share.

Examples:

- **Ammonia (NH_3):** The nitrogen atom in ammonia has a lone pair of electrons that can be donated.
- **Water (H_2O):** The oxygen atom in water has lone pairs of electrons available for donation.

Lewis Acid-Base Reaction:

A Lewis acid-base reaction occurs when a Lewis acid accepts a pair of electrons from a Lewis base, resulting in the formation of a coordinate bond. In this reaction, the base donates the electron pair, and the acid accepts it.

Example Reaction:

- $\mathbf{BF_3 + NH_3 \rightarrow [BF_3 \leftarrow NH_3]}$

In this reaction, BF_3 (a Lewis acid) accepts an electron pair from NH_3 (a Lewis base**)**, forming a complex.

Key Concept Summary:

- **Lewis Acid:** Electron pair acceptor.
- **Lewis Base:** Electron pair donor.
- A **Lewis acid-base reaction** involves the donation of an electron pair from the base to the acid, creating a new bond.

The Lewis theory is more general than the Brønsted-Lowry theory because it can account for reactions that do not involve protons, such as the formation of metal complexes and reactions involving electron-deficient molecules.

Concept of pH and pOH numerical

The concepts of pH and pOH are central to understanding the acidity and alkalinity of aqueous solutions. They are both logarithmic scales based on the concentration of hydrogen ions (H^+) and hydroxide ions (OH^-) in a solution.

pH: pH **is a measure of the acidity or basicity of a solution. It is defined as:**

$$pH = -\log[H^+]$$

where $[H^+]$ is the concentration of hydrogen ions (or protons) in the solution.

- The pH scale ranges from 0 to 14:
  - **pH < 7**: Acidic solution (higher concentration of H^+).
  - **pH = 7**: Neutral solution (equal concentration of H^+ and OH^-).
  - **pH > 7**: Basic (alkaline) solution (lower concentration of H^+ and higher concentration of OH^-).

pOH: pOH **is a measure of the concentration of hydroxide ions (OH^-) in the solution and is defined as:**

$$pOH = -\log[OH^-]$$

where $[OH^-]$ is the concentration of hydroxide ions in the solution.

- The pOH scale also ranges from 0 to 14:
  - **pOH < 7**: Basic solution (higher concentration of OH^-).
  - **pOH = 7**: Neutral solution (equal concentrations of OH^- and H^+).
  - **pOH > 7**: Acidic solution (lower concentration of OH^-).

Relationship Between pH and pOH:

At 25°C, the sum of pH and pOH is always 14, because water autoionizes to produce equal concentrations of H^+ and OH^- ions:

$pH + pOH = 14$

Example 1: Calculating pH from H^+ concentration

Suppose we have a solution with a hydrogen ion concentration $[H^+] = 1\times10^{-3}$M. To find the pH:

$[H^+] = -\log(1\times10^{-3}) = 3$

So, the pH of the solution is 3, which indicates it's acidic.

Example 2: Calculating pOH from OH^- concentration

Suppose we have a solution with a hydroxide ion concentration

$[OH-] = 2\times10^{-5}$ M. To find the pOH:

$pOH = -\log(2\times10^{-5}) \approx 4.7$

Example 3: Given pH, Calculate pOH

If the pH of a solution is 11, we can calculate the pOH:

$pOH = 14 - pH = 14-11 = 3$

So, the pOH is 3, indicating that the solution is basic.

Example 4: Given pOH, Calculate pH

If the pOH of a solution is 5, we can calculate the pH:

pH = 14 – pOH = 14−5 = 9

So, the pH is 9, indicating the solution is basic.

Buffer solutions

Buffer solutions are special types of solutions that resist changes in their pH when small amounts of acid or base are added to them. They are important in many biological and chemical processes, where the pH needs to remain relatively stable. A buffer solution is typically made up of a weak acid and its conjugate base**,** or a weak base and its conjugate acid**.**

Key Concepts:

- **Buffer Capacity**: The ability of a buffer solution to resist changes in pH is called buffer capacity. It depends on the concentration of the acid/base pair. The higher the concentration, the greater the buffer capacity.
- **pH of a Buffer**: The pH of a buffer solution can be calculated using the **Henderson-Hasselbalch equation**:

$$pH = pKa + \log\frac{\text{Conjugate Base}}{\text{Weak Acid}}$$

  - pKa is the negative logarithm of the acid dissociation constant (Ka).
  - The ratio of the conjugate base and the weak acid determines the pH.

Types of Buffer Solutions:

1. **Acidic Buffer**: A buffer solution is typically made up of a weak acid and its salt with strong base. Example: $CH_3COOH + CH_3COONa$
   - **Acetic Acid (CH_3COOH)** is a weak acid.
   - **CH_3COONa** is its salt with strong base. This buffer is commonly used in biological systems, like blood, where maintaining a stable pH is vital for proper enzyme activity.

2. **Basic Buffer:** A buffer solution is typically made up of a weak base and its salt with strong acid. Example: $NH_4OH + NH_4Cl$
    - **NH_4OH** is a weak base.
    - **NH_4Cl** is its salt with strong acid. This buffer is important in many chemical and industrial processes.

Buffer Range:

- A buffer is most effective when the pH is close to the **pKa** of the weak acid/base pair.
- Typically, a buffer works best within **±1 pH unit** of its pKa value.

Buffer solutions are essential in maintaining stable conditions in environments like the human body (e.g., blood), laboratories, and industrial processes.

Buffer solutions used:

Buffer solutions are used in a variety of fields and applications due to their ability to maintain a stable pH. Here are some key uses:

1. Biological Systems (e.g., Blood)

- **Blood pH regulation**: The human blood contains a buffer system (mainly the **carbonic acid-bicarbonate** buffer) that keeps the pH between 7.35 and 7.45. This stable pH is essential for proper cellular function and enzyme activity. If the pH of blood drops too low (acidosis) or rises too high (alkalosis), it can lead to serious health problems.
- **Enzyme function**: Many enzymes function optimally at specific pH values. Buffer solutions help maintain the required pH for enzyme-catalyzed reactions in cells and tissues.

2. Laboratory and Analytical Work

- **Titration experiments**: Buffers are used to maintain a stable pH in titration experiments, ensuring accurate results when measuring the concentration of acids or bases.
- **Calibration of pH meters**: Standard buffer solutions with known pH values (e.g., pH 4.00, 7.00, 10.00) are used to calibrate pH meters in laboratories.
- **Chemical reactions**: Some reactions require a specific pH for proper functioning. Buffers ensure that the pH remains constant during reactions to avoid variations that could affect the outcome.

3. Industrial Applications

- **Food and beverage industry**: Buffers are used in food processing, such as in the preparation of soft drinks and fruit juices, to maintain the desired taste and appearance by controlling the pH.
- **Pharmaceuticals**: Buffer solutions are used in the formulation of medications, especially in intravenous solutions, to ensure that the pH is safe for human administration and doesn't cause tissue irritation.
- **Cosmetics**: Many cosmetic products, like shampoos and lotions, are formulated with buffer solutions to stabilize their pH and ensure the product's effectiveness and safety for skin and hair.

4. **Environmental Science**

- **Water treatment**: Buffers are used in water treatment processes to adjust and maintain the pH of water, ensuring that it is within safe limits for human consumption and aquatic life.
- **Aquariums**: Aquarists use buffer solutions to stabilize the pH of water, which is important for the health of fish and other aquatic organisms.

5. Agriculture

- **Soil pH control**: Buffering agents are used to regulate soil pH for optimal plant growth. Some plants thrive in acidic soils, while others require more neutral or alkaline conditions.
- **Fertilizer applications**: Some fertilizers are buffered to ensure that the nutrients are available to plants at the correct pH.

6. Biotechnology and Molecular Biology

- **PCR (Polymerase Chain Reaction)**: In molecular biology, buffer solutions are used to maintain the pH during PCR, where enzymes like Taq polymerase require specific pH ranges to efficiently replicate DNA.
- **Protein purification**: Buffers are essential for protein extraction and purification processes, as proteins can lose their functionality if the pH is not controlled.

In summary, buffer solutions are vital in any application where maintaining a constant pH is crucial for chemical, biological, or physical processes.

Indicators used in titrations

In titrations, indicators are used to signal the end point of the reaction, which helps in determining the concentration of the unknown solution. The indicator undergoes a visible change, usually a color shift, when the reaction has reached the equivalence point. Different types of indicators are used depending on the nature of the titration (acid-base, redox, complexometric, etc.). Here are some common indicators for acid-base titrations:

1. Phenolphthalein

- **Used for**: Strong acid vs. strong base or weak acid vs. strong base.
- **Color change**: Colorless in acidic solution, pink in basic solution (pH range ~8.3 to 10.0).
- **Usage**: It is a common indicator for titrations involving strong bases like NaOH.

2. Methyl Orange

- **Used for**: Strong acid vs. strong base or Strong acid vs. weak base.
- **Color change**: Red in acidic solution, yellow in basic solution (pH range ~3.4 to 4.4).
- **Usage**: Used when titrating with strong acids like HCl and weak bases like ammonia.

3. Bromothymol Blue

- **Used for**: Strong acid vs. strong base titrations.
- **Color change**: Yellow in acidic solution, blue in basic solution (pH range ~6.0 to 7.6).
- **Usage**: Commonly used for titrations involving NaOH and HCl.

4. Methyl Red

- **Used for**: Strong acid vs. weak base titrations.
- **Color change**: Red in acidic solution, yellow in basic solution (pH range ~4.4 to 6.2).
- **Usage**: Often used in titrations involving acids like HCl and weak bases like ammonia.

5. Thymolphthalein

- **Used for**: Weak acid vs. strong base titrations.
- **Color change**: Colorless in acidic solution, blue in basic solution (pH range ~9.3 to 10.5).
- **Usage**: Often used for titrations where the pH goes from acidic to basic.

6. Universal Indicator

- **Used for**: General acid-base titrations.

- **Color change**: Changes color through a broad range (from red in strong acid to purple in strong base) covering pH 4-10.
- **Usage**: Useful in experiments where a broader pH range is needed.

7. Alizarin Yellow R

- **Used for**: Weak acid vs. strong base.
- **Color change**: Yellow in acidic solution, red in basic solution (pH range ~10.1 to 12.1).
- **Usage**: Ideal for titrations where the pH will reach higher basic values.

8. Indicator for Redox Titrations:

- **Potassium Dichromate**: Used with indicators such as diphenylamine.
- **Color change**: Can change color as a result of redox reactions.

These indicators are chosen based on the pH at the equivalence point of the titration, ensuring that their color change happens near that point for accurate results.

The solubility product

The **solubility product** (often denoted as Ksp) is a constant that represents the solubility of a sparingly soluble salt in water. It is used to describe the equilibrium between a solid and its ions in a saturated solution.

In simpler terms, it tells us how much of a substance can dissolve in water to form a solution. The higher the value of Ksp, the more soluble the compound is.

For example, if you have a salt like calcium fluoride (CaF_2):

$$CaF_2(s) \rightleftharpoons Ca^{2+}(aq) + 2F^-(aq)$$

The solubility product would be:

$Ksp = [Ca^{2+}] [F^-]^2$

Where:

- $[Ca^{2+}]$ is the concentration of calcium ions in the solution.
- $[F^-]$ is the concentration of fluoride ions.

The solubility product is a way of quantifying how much of the solid will dissolve into the solution at equilibrium.

If you know the concentrations of the ions in the solution, you can use Ksp to predict whether a solution will be saturated or if a precipitate will form.

Common ion effect with their applications

Common ion effect is a phenomenon in which degree of dissociation of any weak electrolyte is supressed by addition of small amount of electrolyte containing a common ion.

Example:

- When sodium acetate is added in acetic acid solution then ionization (dissociation) of acetic acid decreases due to increased concentration of acetate ion (common ion)

  $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$
  Acetic acid Acetate ion

  $CH_3COONa \rightarrow CH_3COO^- + Na^+$
  Sodium acetate Acetate ion

- When ammonium chloride is added in ammonium hydroxide solution then ionization (dissociation) of ammonium hydroxide decreases due to increased concentration of ammonium ion (common ion).

  $NH_4OH \rightleftharpoons NH_4^+ + OH^-$
  Ammonium hydroxide Ammonium ion

  $NH_4Cl \rightarrow NH_4^+ + Cl^-$
  Ammonium chloride Ammonium ion

The **common ion effect** refers to the shift in equilibrium that occurs when a solution already contains one of the ions involved in a reversible reaction, and additional amounts of that ion are added to the system. This effect reduces the solubility of salts and affects the position of equilibrium in certain reactions.

Basic Explanation:

- When you add an ion that is already part of the equilibrium, it causes the reaction to shift in a way that reduces the concentration of that ion by either forming more solid (precipitating) or decreasing the dissociation of a substance.

For example, if you add NaCl to a solution of **AgCl**, the common ion is **Cl^-**, which reduces the solubility of **AgCl** (because it increases the concentration of Cl^-, making the system reach equilibrium sooner by precipitating more silver chloride).

Applications of Common Ion Effect:

1. **Solubility of Salt:**
   - The solubility of a salt like **AgCl** decreases if a solution already contains a common ion, such as Cl^- (from NaCl). The more Cl^- you add, the less **AgCl** will dissolve.
2. **Buffer Solutions:**
   - In a buffer solution (a solution that resists changes in pH), adding a common ion helps to maintain the pH by neutralizing added acids or bases. For example, a mixture of **acetic acid** and **sodium acetate** forms a buffer, where **acetate** is the common ion.
3. **Precipitation Reactions:**
   - In qualitative analysis, the common ion effect is used to selectively precipitate certain ions from a solution. For example, if you want to precipitate calcium as calcium carbonate, adding a common ion (like $\mathbf{CO_3^{2-}}$ from Na_2CO_3) will reduce the solubility of $\mathbf{CaCO_3}$.
4. **Acid-Base Titration:**
   - In titrations, the common ion effect can help to slow down the reaction between a strong acid/base and a weak acid/base by shifting the equilibrium. For example, adding a salt that shares a common ion with a weak acid can reduce the dissociation of the acid.

In Summary:

The **common ion effect** is crucial in controlling the solubility of salts, making buffer solutions, selectively precipitating ions, and managing reactions in titrations. It helps in predicting and controlling chemical behavior in various systems.